European Polymer Journal 219 (2024) 113391

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

A highly efficient oxygen tolerant visible-light mediated RAFT polymerization

Chengyu Zhou [a,1], Lingfeng Xia [a,1], Yan Xiong [a,*], Chao Weng [b,*], Yufeng Xiao [a], Xuewen Wu [a,*]

[a] Department of Pharmaceutical Engineering, School of Chemical Engineering, Xiangtan University, Xiangtan 411105, PR China
[b] Key Laboratory of Environmentally Friendly Chemistry and Applications of Ministry of Education, College of Chemistry, Xiangtan University, Xiangtan 411105, PR China

ARTICLE INFO

Keywords:
Oxygen Tolerance
RAFT Polymerization
Acrylates
Acrylamides
Glycol ethers
Green synthesis

ABSTRACT

Recently, oxygen-tolerant reversible addition-fragmentation chain transfer (RAFT) polymerization has attracted increasing attention in polymer chemistry. Inspired by the monomers containing glycol ethers group have high polymerization rates and even achieved a desirable oxygen tolerance. Herein, we proposed a simple, effective, low-cost, and eco-friendly strategy to accomplish the oxygen-tolerant visible-light activated RAFT polymerization of acrylates and acrylamides in a mixed solution of glycol ether and water at ambient temperature. This photo-mediated RAFT polymerization does not require additives, prior deoxygenation, or inert gas protection. The acrylates and acrylamides with high conversion under irradiation for 2 h, and the resulting polymers were narrow unimodal and narrow molecular weight distributions (PDI<1.3). In addition, the "living" and well-controlled behaviors of this oxygen-tolerant photo-mediated RAFT polymerization were demonstrated through water content in solvents, headspace, initial concentration of monomers, various monomers, chain lengths, light ON/OFF experiment, and high-end group fidelity of block copolymers. This RAFT polymerization also opens up a wide avenue for the industrial production of functional polymeric materials via an *in situ* chain extension strategy. These results indicate that the mixed solution of glycol ethers and water are effective green media for promoting photo-mediated RAFT polymerization without additives or tedious degassing processes.

1. Introduction

Controlled/living radical polymerization (CLRP), including reversible addition-fragmentation chain-transfer (RAFT), atom-transfer radical polymerization (ATRP), and nitroxide-mediated polymerization (NMP), provides a facile synthesis of the targeted polymers with a predetermined molecular weight, low dispersity, and chain-end functionality.[1–4] Among them, RAFT has been implemented in various fields of commercial adhesive gules, coating, ink materials for printing, 3D printing, and surfactants.[5–7] More recently, visible-light mediated RAFT polymerization-induced self-assembly (RAFT-PISA) has been regarded as one of the versatile approaches to construct precisely controlled high solids content (up to 50 % w/w) multi-block polymers.[8,9] It is well known that the principal advantages of photo-controlled RAFT polymerization are the ability to achieve time and space-controlled polymer materials under mild conditions,[3] whereas the fatal disadvantage of photo-controlled RAFT polymerization, like other CLRP, is its sensitivity to molecular oxygen.

In recent years, oxygen-tolerant RAFT polymerization has been considered to overcome the drawback of oxygen sensitivity to a certain extent and is promising for the industrial-scale production of functional polymeric materials and biomaterials.[4,6,10,11] In terms of the photo-CLRP mechanism, most of the polymer chains are in the dormant state during the polymerization processes, and only a small amount of them are active as chain propagation radicals.[12] Therefore, the propensity of the propagating radical intermediate to react with triplet oxygen and form peroxygen radicals or hydroperoxides makes it extremely challenging for industrial applications.[13] Generally, complex physical or chemical deoxygenation strategies are required to overcome the oxygen-inhibit phenomenon. Removing oxygen through physical displacements, such as inert gas purging, can effectively terminate the inhibition of molecular oxygen on CLRP. However, the physical deoxygenation process is time-consuming, cumbersome, and requires specialized equipment. Furthermore, chemical approaches, such as introducing reducing

* Corresponding authors.
E-mail addresses: Xiongyan@xtu.edu.cn (Y. Xiong), wengc2013@163.com (C. Weng), wxw@xtu.edu.cn (X. Wu).
[1] These authors contributed equally to this work.

https://doi.org/10.1016/j.eurpolymj.2024.113391
Received 17 June 2024; Received in revised form 14 August 2024; Accepted 17 August 2024
Available online 18 August 2024
0014-3057/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

agents to scrub molecular oxygen, consuming sufficient dissolved oxygen, and converting molecular oxygen into reactive oxygen species, provide efficient means to dispel oxygen in the polymerization solvents. [14] Typically, the oxygen tolerant photo mediated RAFT polymerization is achieved by adding zinc tetraphe nylporphyrine (ZnTPP)[15,16], small molecule dyes[17–19], hyperconjugated systems[20,21], organoboron compounds[22–24], quantum dots[25–27], vitamins[28], and enzymes[29–33] into the polymerization system, which consume molecular oxygen or convert them to other non-interfering substances. However, these enzymes and other additives could be difficult to remove from resulting polymers and contaminate final materials. Therefore, there is an urgent need to develop simpler, more effective and eco-friendly strategies for the oxygen tolerant photo mediated RAFT polymerization in ambient conditions.

Recently, it has been found that polymeric monomers containing glycol ethers group are relatively insensitive to oxygen inhibition, have a high polymerization rate and even achieve oxygen tolerant polymerization process [19,34–38]. Rostami's group found that glycol ether monomers have superior oxygen tolerance[34]. Cheng and coauthors achieved oxygen-tolerant polymerization of poly(ethylene glycol) methacrylate (PEGMA) in a continuous reactor.[35] Boyer's group observed that a higher mole fraction of polyethylene glycol diacrylate (PEGDA) accelerates the light-controlled RAFT polymerization process. [19] Shi and coauthors implemented ultra-fast, "living" and well-controlled RAFT polymerization of monomer PEGMA in mild aqueous conditions, yielding high monomer conversions (>80 %).[37] Interestingly, 2-[2-(2-methoxyethoxy) ethoxy] acetic acid (TODA) has three glycol ether groups and is thought to remove molecular oxygen and react with peroxyl radicals to generate oxidation products. [39] Hence, we hypothesized that the solvents containing glycol ether groups may be fascinating solvents for photo-activated RAFT polymerization of acrylate and acrylamide monomers.

Herein, we proposed a simple, effective, low-cost, and eco-friendly method to perform oxygen-tolerant RAFT polymerization of acrylates and acrylamides in glycol ethers solvents without additives, prior deoxygenation, or inert gas protection. In the present study, n-butyl acrylate (nBA), methyl acrylate (MA), and PEGMA were chosen as the typical acrylate monomers and N-isopropyl acrylamide (NIPAM), N, N-dimethyl acrylamide (DMAA), diacetoneacrylamide (DAAm) was chosen as the typical acrylamide monomers. 2-methoxyethanol (MOE), 1, 2-dimethoxyethane (DMOE), and 2-butoxyethanol (BOE) were selected as the glycol ethers model solvents. 2, 4, 6-Trimethylbenzoyldiphenylphosphine oxide (TPO), S-1-dodecyl-S- (α,α''-dimethyl-α''-acetic acid) trithiocarbonate (DDMAT) was used as the photoinitiator, RAFT chain transfer agent (CTA), respectively.

2. Results and discussion

2.1. Dissolved oxygen

Oxygen dissolved in the solvent is known to inhibit RAFT polymerization processes. Therefore, it is crucial to investigate the dissolved oxygen consumption in the polymerization solvent in the presence of an initiator. In this study, the dissolved oxygen consumptions in MOE and MOE/water under an air environment were first investigated using a dissolved oxygen meter (Fig. 1a). The trends of photo-initiators TPO and RAFT agent DDMAT under 405 nm visible light LED arrays were also monitored using UV–visible absorption spectroscopy (Fig. S1). As shown

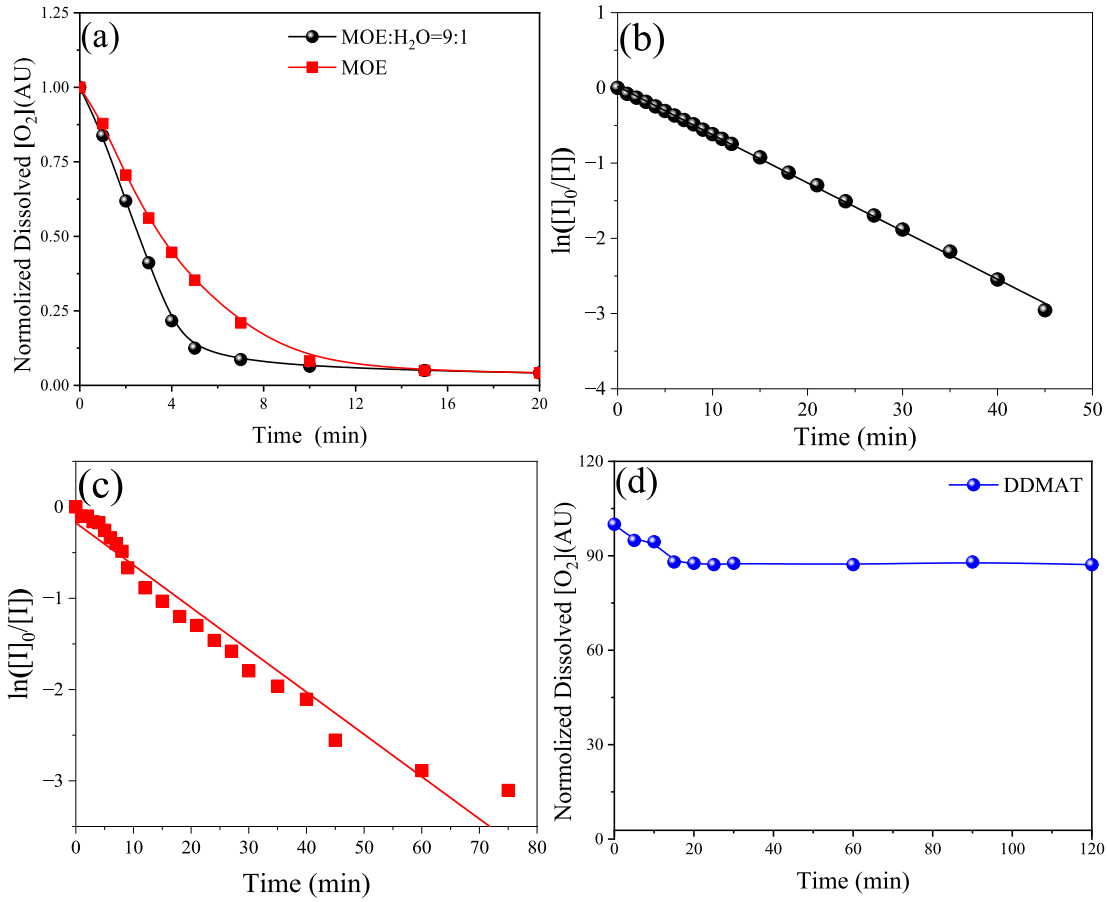


Fig. 1. The dissolved oxygen consumption in MOE and MOE/H_2O=9/1 (v/v) at present of TPO (0.1 mmol/L) under a 405 nm LED array (a); quasi-primary kinetic model fitting for TPO degradation in (b) MOE/H_2O=9/1 (v/v) and (c) MOE; The dissolved oxygen consumption in MOE/H_2O=9/1 (v/v) at present of DDMAT (0.1 mmol/L) under a 405 nm LED array (d).

in Fig. 1a, under the 405 nm visible LED array, TPO consumes dissolved oxygen in both MOE and MOE/water solvents, which is significantly faster in MOE/water than in MOE solvent.

In general, the decomposition of photo-initiators ($I \rightarrow 2R\bullet$) can be described by the pseudo-first-order kinetic equation (1):

$$R_d = -\frac{d[I]}{dt} = k_d[I] \tag{1}$$

where the negative sign indicates that the initiator concentration [I] decreases with time t and k_d (min^{-1}) is the decomposition rate constant.

According to the kinetic equation (1), the rate equation for the decomposition of the photo-initiator is equation (2),

$$ln\frac{[I]_0}{[I]} = -k_d t \tag{2}$$

where $[I]_0$, [I], k_d, t represents the photo-initiator concentration (mmol/L), the instantaneous concentration, decomposition rate constant, and time, respectively.

Fig. 1b and Fig. 1c (also seen in Fig. S1a and Fig. S1b) showed that the free radicals generated by TPO were consistent with the pseudo-first-order kinetic model in MOE and MOE/water solvents. The apparent decomposition rate constants k_d for MOE/water and MOE were 0.064 min^{-1} and 0.046 min^{-1} with a half-life of 11 min and 16 min, respectively. This indicates that the water in MOE accelerates the consumption of dissolved oxygen and the generation of free radicals. Fig. 1d revealed that the RAFT agent DDMAT has a negligible ability to deplete dissolved oxygen in MOE/water solvents. In addition, Fig. S1c showed the excellent photostability of DDMAT under 405 nm LED array irradiation monitored by a UV–visible spectrophotometer in MOE/water = 9/1 solvent. The loss of DDMAT was found to be negligible after 180 min of irradiation. Therefore, it is reasonable to hypothesize that oxygen-tolerant RAFT polymerization can be achieved in a mixed solution of glycol ether and water.

2.2. Polymerization solvent

Therefore, we further explored the oxygen-tolerant visible light-mediated RAFT polymerization using glycol ethers as the polymerization solvent. In the present study, dimethyl sulfoxide (DMSO) was selected as the control solvent because it was a typical solvent for oxygen-tolerant photo-controlled RAFT polymerization.

The effects of different solvents on photo-mediated polymerization were summarized in Table 1, Table S1 and Fig. S2. The monomer conversions were low (<5%) when irradiated with 405 nm LED arrays in pure organic solvent for 2 h at ambient temperature (Table S1, entries 1—5). Interestingly, considerable conversions (>90 %) and excellent polydispersity indices (PDI, M_w/M_n) (Đ < 1.20) were obtained after introducing small amounts of water into the glycol ethers solvents (Table 1, entries 1—4). As shown in Fig. S2, the GPC traces of the resulting polymers were narrow unimodal curves, which demonstrated a well-controlled performance of this photo-mediated RAFT polymerization in a mixed solution of glycol ether and water. In addition, Table 1 also showed that the polymerization of nBA with DMSO/water in sealed flasks produced polymer with lower conversion (16.3 %) after 2 h irradiation (Table 1, entry 5). The results suggested that the water and glycol ethers should play a key role in this photo-controlled RAFT polymerization. Therefore, the mixed solution of glycol ether and water may be an ideal solvent for this oxygen-tolerant visible light-mediated RAFT polymerization.

Table 1
Results of the visible-light mediated RAFT polymerization in different aqueous solvents. [a]

Entry	Monomer	Solvent	Conv%	Mn (kg/mol)	Đ
1	nBA	MOE	94.1	10.95	1.18
2	MA	MOE	94.4	11.09	1.06
3	NIPAM	DMOE	93.9	11.07	1.12
4	nBA	BOE	97.8	12.90	1.20
5	nBA	DMSO	16.5	2.48	1.10

[a] The molar ratio of $[monomer]_0$: $[DDMAT]_0$: $[TPO]_0$ was 100:1:0.5, the solids content of monomer was 25 wt% and the irradiation time was 2 h, using [organic solvent]/water = 9/1 v/v as the solvents.

As shown in Fig. 2a and Table S2, the nBA obtained high conversions (>91 %) and excellent PDIs (Đ < 1.30) under 2 h irradiation, indicating the water in the MOE provided the favorite apparent polymerization rate. Moreover, the GPC traces of these polymers (Table S2, entries 3—4) were narrow unimodal curves after 2 h of visible light irradiation, indicating that this visible light-mediated RAFT polymerization reaction performed well in MOE solvents containing appropriate water. In addition, the obvious shoulder peaks at the high molecular weight side in GPC traces were observed when the water content in the MOE solvent was lower than 5 % (v/v) (Table S2, entries 1—2) or higher than 12.5 % (Table S2, entries 5—6). The equilibrium constant of CLRP in water leads to a high concentration of free radicals, which may result in dead chains[2], such as the occurrence of radical–radical termination or chain transfer to the solvent. Therefore, the results also indicated that the glycol ethers containing appropriate water were the ideal media for this RAFT polymerization.

2.3. Headspace volume

To investigate the effect of air on this photo-mediated RAFT polymerization, nBA was chosen as the typical monomer. As shown in Fig. 2b. and Table S3, performing this RAFT polymerization under air in a 19 ml flask, the conversions of nBA were lower than 5 % (Table S3, entry 1). The broad and asymmetrical GPC trace ($M_{n,}$ = 0.53 kg mol^{-1}, Đ = 1.16) also indicated that this polymerization of nBA was uncontrollable under large headspace (Fig. 2b). Further reduction of the headspace, high monomer conversions (>95 %) were obtained under 2 h of visible light irradiation, yielding polymers with low PDIs (Đ < 1.20) (Table S3, entries 2—5) and narrow unimodal GPC traces (Fig. 2b). Hence, we demonstrated the reasonable oxygen tolerance of this photo-mediated RAFT polymerization under appropriate headspace volume.

2.4. Solids content

As is well known, the concentration of monomers not only affects the polymerization rate but also determines the free radical flow rate, which requires a certain threshold for successful polymerization reactions. Therefore, the initial concentration of nBA effects on this RAFT polymerization was further investigated. The low initiation efficiency was observed at the low initial concentration of monomer. Table S4 indicated that when the initial concentrations of nBA were below 20 wt%, the conversions of nBA were below 25 wt% (Table S4, entries 1—2), and the GPC traces of the resulting polymers were unsatisfactory (Fig. 2c). It was observed that the theoretical molecular weight ($M_{n,\ th}$ = 1.09 kg mol^{-1}) and the experimental molecular weight ($M_{n,\ GPC}$=0.52 kg mol^{-1}) deviated considerably (Table S4, entries 1). When the initial concentrations of nBA exceeded 20 %, the conversions of nBA were high (>90 %) (Table S4, entries 3—6). In general, the theoretical molecular weight ($M_{n,\ th}$ = 12.00—12.90 kg mol^{-1}) and the experimental molecular weight ($M_{n,\ GPC}$=10.95—13.71 kg mol^{-1}) are in good agreement (Table S4, entries 3—6). The PnBA acquired perfect PDIs (Đ < 1.20) under irradiation for 2 h (Table S4, entries 1–6), suggesting a wide concentration of monomers was suitable for this visible light-mediated RAFT polymerization. In addition, these GPC traces (Fig. 2c) were quite narrow and unimodal, suggesting a well-controlled performance of this photo-mediated RAFT polymerization at a higher initial concentration of monomers.

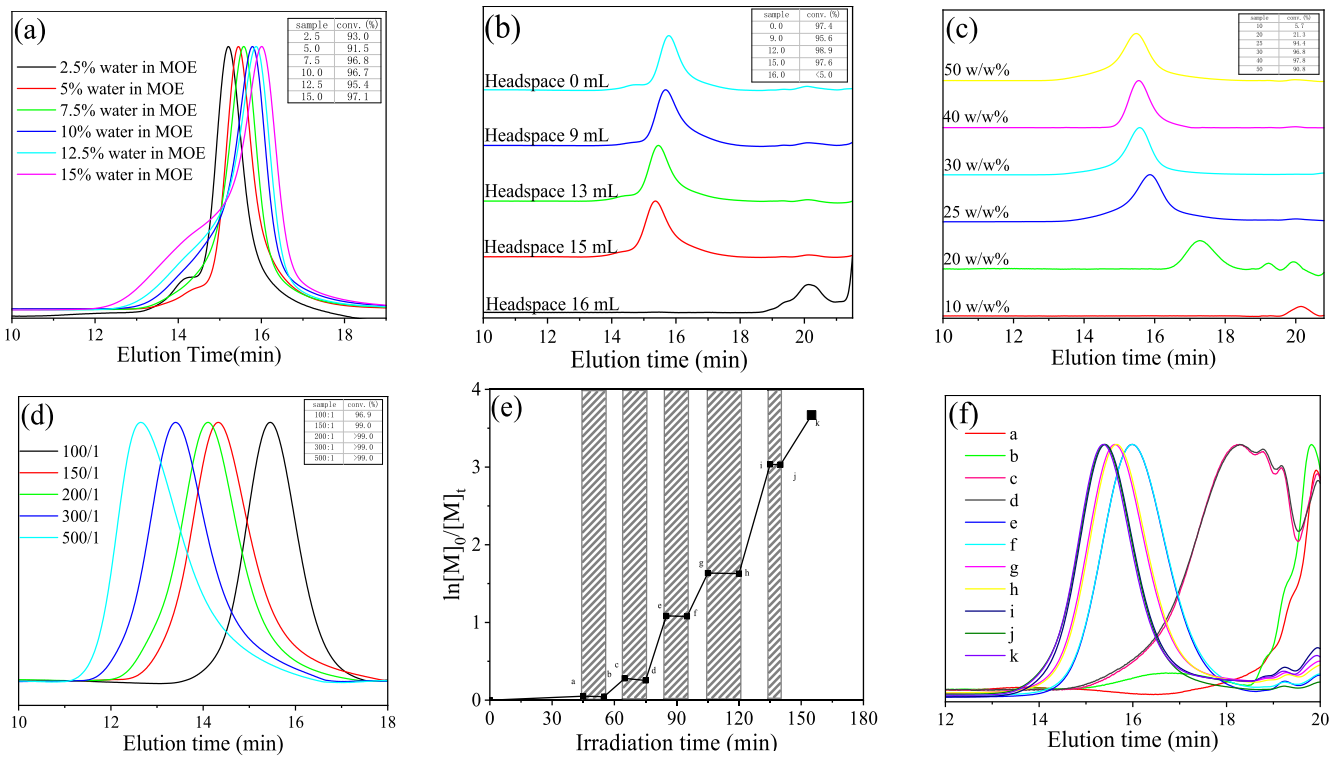


Fig. 2. The effects of water (a), headspace (b), monomer content (c), and degrees of polymerization (d) on this RAFT polymerization in a mixed solution of MOE and water solvents. Polymerization conditions for (a)- (c): $[nBA]_0$:$[DDMAT]_0$:$[TPO]_0$ = 100: 1: 0.5, Polymerization conditions for (d): $[DAAm]_0$:$[DDMAT]_0$ from 100:1 to 500:1, under irradiation with 405 nm LEDs light ($I_{420\ nm}$ = 200 μW cm^{-2}) at ambient temperature for 2 h. Light ON/OFF experiment (e) and the GPC traces (f) at different times. Polymerization conditions for (e) and (f), the concentration of DAAm was 30 w/w %. The feed molar ratio of $[DAAm]_0$:$[DDMAT]_0$:$[TPO]_0$ was 100:1:0.5.

2.5. The degrees of polymerization

The diverse degrees of polymerization (DPs) were also constructed using diacetone acrylamide (DAAm) as a typical monomer to assess the applicability of this photo-mediated RAFT polymerization (Fig. 2d and Table S5). As shown in Fig. 2d, increasing the feed molar ratio of $[DAAm]_0$: $[DDMAT]_0$: $[TPO]_0$ from 100: 1: 0.5 to 500: 1: 0.5, high monomer conversions (>96 %) were obtained under 2 h of visible light irradiation, yielding polymers with desirable PDIs (Đ < 1.32), excellent control over M_W (Table S5, entries 1—5) and narrow unimodal GPC traces, suggesting that this visible-light-activated RAFT polymerization has well-controlled characteristics at different DPs. Table S5 also revealed that the M_n of obtained PDAAm polymers increased proportionally with the feed molar ratio of DAAm. These further confirmed the well-controlled behaviors of this RAFT polymerization.

2.6. Light ON/OFF experiment

A favorable characteristic of visible light-activated RAFT polymerization is to control temporally by light ON/OFF experiment. As shown in Fig. 2e and 2f, this photo-mediated RAFT polymerization could be activated after around a 30 min induction period without prior deoxygenation. The conversions of DAAm increased merely during light ON states and no polymerization reaction was observed during light OFF states (Fig. 2e and Table S6), indicating that the visible light-activated RAFT polymerization could be manipulated by an ON/OFF experiment. In addition, the molecular weights of NMR and GPC were in excellent agreement (Table S6), indicating minimal termination. The GPC traces (Fig. 2e, 2f) of the corresponding samples also revealed that the molecular weight of the polymers only increased during the light ON state, proving the fidelity of the end-group, which further demonstrated the photo-controlled capability of our RAFT protocol.

2.7. Kinetic studies of photo-mediated RAFT polymerization

The typical acrylate and acrylamide monomers, DAAm and nBA, were used to further explore the applicability of this photo-mediated RAFT polymerization under the N_2 atmosphere and air atmosphere. In general, the oxygen tolerance RAFT polymerization was performed air atmosphere (25 mL of reaction solution and about 13 mL of headspace of air in a sealed flash). For the visible light-mediated polymerization system under the N_2 atmosphere, the mixture was sealed with a rubber septum, degassed for 30 min, and bubbled with high-purity N_2 for another 30 min.

There are good linear relationships between $\ln([M]_0/[M]_t)$ and irradiation time (Fig. 3a), and the apparent polymerization rate constants (k_p^{app}) are 1.77×10^{-2} min^{-1} and 1.37×10^{-2} min^{-1}, respectively, indicating that the polymerization characteristics are similar in air and N_2 atmosphere, both of which are consistent with pseudo-first-order kinetics. In addition, the polymerization under air showed a longer induction time (34 min) than that of under N_2 atmosphere (10 min). The longer induction period suggested that this oxygen-tolerant photo-mediated RAFT polymerization was initiated only after the complete consumption of dissolved oxygen in the air atmosphere. Furthermore, the CTA end group in the polymer was confirmed by ^{1}H NMR (Fig. S3), which further supported the livingness of this oxygen-tolerant visible light-activated RAFT polymerization. GPC traces (Fig. S4) showed narrow unimodal curves indicative of well-controlled polymerizations in these systems. Fig. 3b and Fig. 3c (also seen Table S7 and Table S8) revealed that the measured M_n increased linearly with the conversion of nBA, whereas a low Đ (<1.20) was maintained throughout polymerization, indicating the living nature of photoinitiated polymerization under air and N_2 atmosphere. However, we found that the deviation of the experimental M_w obtained by GPC from the theoretical M_w of PnBA was larger in the air atmosphere than in the N_2 atmosphere as the

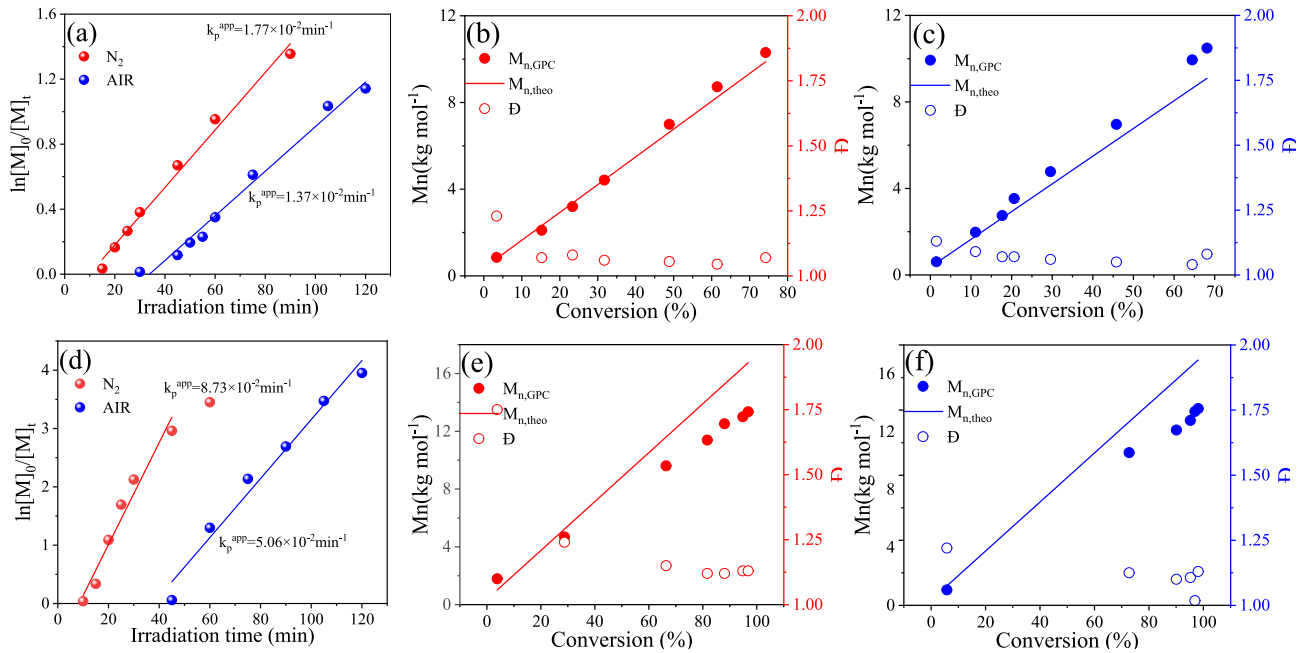


Fig. 3. Photo-mediated RAFT polymerization of nBA and DAAm at ambient temperature under 405 nm LEDs ($I_{420\ nm}$ = 200 μW cm^{-2}). The first-order kinetic plot of $\ln[M]_0/[M]_t$ for nBA under N_2 and air atmosphere *vs* irradiation time(a). M_n (filled circle) and M_w/M_n (open circle) *vs* conversions under N_2 (b) and air atmosphere (c). The first-order kinetic plot of $\ln[M]_0/[M]_t$ for DAAm under N_2 and air atmosphere *vs* irradiation time (d). M_n (filled circle) and Mw/Mn (open circle) *vs* conversions under N_2 (e) and air atmosphere (f). The concentrations of nBA and DAAm were 30 w/w %, respectively. The feed molar ratio of $[Monomer]_0$: $[DDMAT]_0$: $[TPO]_0$ was 100: 1: 0.5.

irradiation time increased. This may be because the RAFT chain transfer agent is partially consumed by oxygen during the reaction in the atmosphere.

To further prove the universality of this visible light-mediated RAFT polymerization, we also investigated the photo-initiated polymerization of water-soluble DAAm under air and N_2 atmosphere. As shown in Fig. 3d, the k_p^{app} of DAAm was 8.73 × 10^{-2} min^{-1} under N_2 and 5.06 × 10^{-2} min^{-1} under air atmosphere, respectively. Fig. 3d also displayed that the polymerization induction period (38 min) under the air atmosphere was longer than that under the N_2 atmosphere (8 min). It is clear that molecular oxygen dissolved in the solvent retards the polymerization kinetics (induction period, polymerization reaction rate), which is considered to be an inherent feature of oxygen-tolerant RAFT polymerization.[6] Fig. 3e and Fig. 3f (also seen Table S9 and Table S10) indicated that the experimental M_n increased linearly with irradiation time. ^1H NMR also confirmed the CTA end-groups in the polymer (Fig. S5), which further supported the livingness of this polymerization in a mixed solution of glycol ether and water. GPC traces (Fig. S6) show narrow unimodal curves, which also indicates that this visible light-mediated RAFT polymerization process is well controlled. These also indicate that the polymerization behavior of DAAm in the air is similar to that in the N_2 atmosphere.

Therefore, this oxygen-tolerant photo-mediated RAFT polymerization of typical acrylate and acrylamide monomers demonstrated similar "living" and well-controlled behaviors. However, as the polymerization proceeded, the experimental M_n was about 25 % lower than the theoretical molecular weight, suggesting that the rapidly continuous generation of radicals, led to the initiation of new chains resulting in shorter PDAAm-DDMAT. It also indicated that an excessive reaction rate is not conducive to controlling the polymerization process.

2.8. One-pot in situ chain extension strategy

More recently, it has been growingly attractive and challenging to explore a simple and convenient strategy for multi-block copolymer materials. Therefore, we attempted to further confirm the livingness of this oxygen-tolerant visible light-activated RAFT polymerization and the chain-end fidelity in a mixed solution of glycol ether and water *via* a one-pot chain extension protocol. The developed *in situ* chain extension strategy used DAAm as the initial monomer with a monomer solids content of 30 w/w%. Further, the macromolecular CTA was used directly after polymerization without purification, and then new monomers were added to extend the existing chains. Consequently, the ABAB and ABCD tetra block copolymers were successfully constructed through visible light-activated RAFT polymerization (Fig. 4). In the synthesis of tetra block copolymers, nBA, DMAA, and PEGMA were used as the second, third and fourth monomers, respectively. As shown in Fig. 4, the GPC traces of ABAB (also seen in Fig. S7 and Table S11) and ABCD (also seen in Fig. S8 and Table S12) tetra block copolymers peaks shift to higher molecular weight after each chain extension. The blocks of copolymers exhibited narrow unimodal curves (Đ < 1.30). Hence, this oxygen-tolerant visible light-activated RAFT polymerization provided an attractive avenue for multifunctional materials.

2.9. Mechanism

A possible polymerization mechanism was described in Scheme 1 according to the above experimental results. TPO underwent photolysis under the irradiation of visible light LED arrays, generating reactive radicals in a mixed solution of glycol ether and water, which then interacted with CTA to generate carbon-centered or sulfur-centered free radicals. As a result, the free radicals consume the trace molecular oxygen in the polymerization system to form some oxidation products. Water and glycol ethers accelerate the generation of oxidation products and the consumption of dissolved oxygen in this photo-mediated RAFT polymerization. Interestingly, the experimental molecular weights of acrylate monomers polymerized in an air atmosphere deviated negatively from the theoretical molecular weights, confirming the kinetic ($k_0 > k_1 > k_2$) tendency in this polymerization process (Fig. 3c). In addition, the polymerization of acrylamide monomers suggests that DDMAT also affects the polymerization process, which is related to not only the matching between CTA and monomers but also the kinetic of the

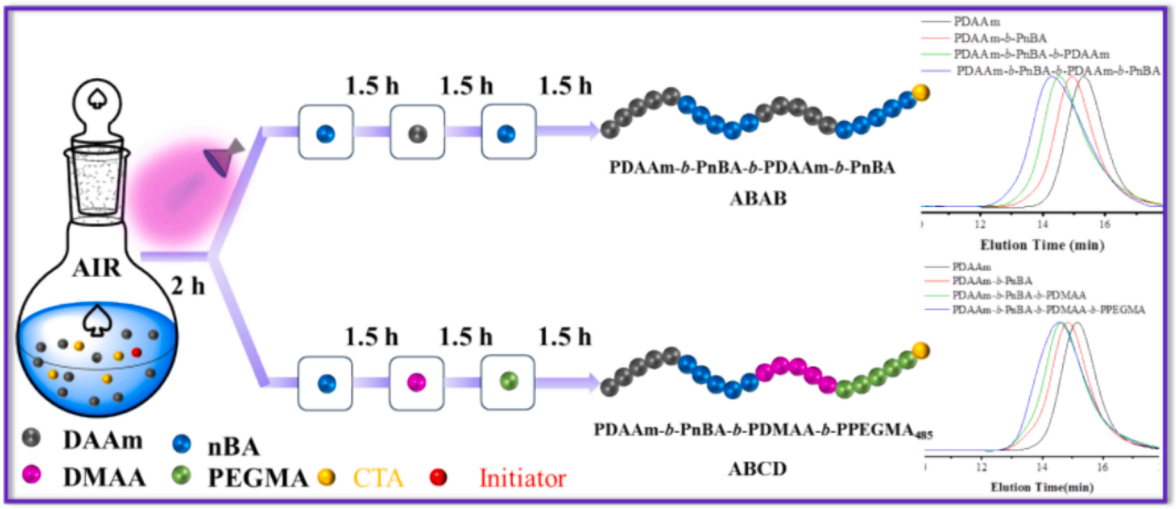
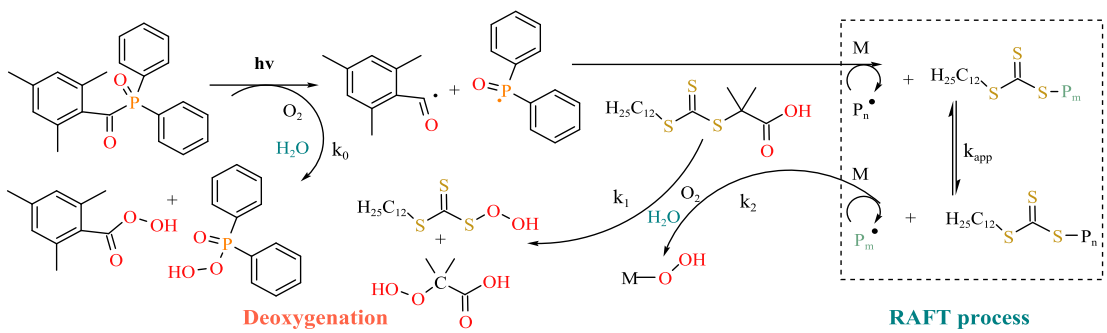


Fig. 4. One-pot synthesis of multiblock polymers via photo-mediated RAFT polymerization in aqueous media at ambient temperature. The GPC traces of ABAB (up) and ABCD (down) copolymers.



Scheme 1. Plausible polymerization mechanism in a mixed solution of glycol ether and water.

oxidation products ($k_0 > k_2 > k_1$).

3. Conclusion

In summary, we developed a green, low-cost oxygen tolerant visible light-mediated RAFT polymerization for acrylate and acrylamide monomers in a mixed solution of glycol ether and water at ambient temperature. We also demonstrated the controlled/living behaviors of this RAFT polymerization. The acrylate and acrylamide monomers achieved high conversions within 2 h and a molecular weight dispersion of Đ < 1.30. In addition, good temporal control over this oxygen-tolerant photo-mediated RAFT polymerization could be achieved through a Light ON/OFF experiment. Finally, the high chain-end fidelity of the polymers was demonstrated via a one-pot chain extension strategy. Importantly, this RAFT polymerization can be carried out without additives or prior deoxygenation, reducing technician and equipment requirements, and thus saving costs. Therefore, the developed RAFT polymerization affords a bright avenue in the industrial production of functional materials.

CRediT authorship contribution statement

Chengyu Zhou: Writing – review & editing, Software, Methodology, Investigation, Conceptualization. **Lingfeng Xia:** Writing – review & editing, Visualization, Resources, Methodology, Data curation, Conceptualization. **Yan Xiong:** Writing – review & editing, Validation, Supervision, Methodology, Data curation. **Chao Weng:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization. **Yufeng Xiao:** Writing – review & editing, Methodology, Data curation. **Xuewen Wu:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the Natural Science Foundation of Hunan Province, China (2021JJ30667), the Scientific Research Foundation of Hunan Provincial Education Department, China (22A0107), and the Hunan Province College Students' Innovation and Entrepreneurship Training Program, China (S202310530079).

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113391.

References

[1] X. Li, Y. Huang, W.F. Wei, W.L. Guo, Z. Luo, J. Xu, T. Cai, Metalloporphyrin-anchored 2D MOF nanosheets as highly accessible heterogeneous photocatalysts towards cytocompatible living radical polymerization, Chem. Eng. J. 434 (2022) 134692.

[2] K. Kapil, A.M. Jazani, G. Szczepaniak, H. Murata, M. Olszewski, K. Matyjaszewski, Fully oxygen-tolerant visible-light-induced ATRP of acrylates in water: Toward synthesis of protein-polymer hybrids, Macromolecules (2023) 2017–2026.

[3] Z. Wu, W. Fang, C. Wu, N. Corrigan, T. Zhang, S. Xu, C. Boyer, An aqueous photo-controlled polymerization under NIR wavelengths: Synthesis of polymeric nanoparticles through thick barriers, Chem. Sci. 13 (39) (2022) 11519–11532.

[4] J. Yeow, R. Chapman, A.J. Gormley, C. Boyer, Up in the air: oxygen tolerance in controlled/living radical polymerisation, Chem. Soc. Rev. 47 (12) (2018) 4357–4387.

[5] B. Zhao, J. Li, Y. Xiu, X. Pan, Z. Zhang, J. Zhu, Xanthate-based photoiniferter RAFT polymerization toward oxygen-tolerant and rapid living 3D printing, Macromolecules 55 (5) (2022) 1620–1628.

[6] Y. Lee, C. Boyer, M.S. Kwon, Photocontrolled RAFT polymerization: Past, present, and future, Chem. Soc. Rev. 52 (9) (2023) 3035–3097.

[7] V.A. Bobrin, H.G. Hackbarth, J.O. Bonsu, Y. Yao, N.M. Bedford, D. Kundu, J. Zhang, N. Corrigan, C. Boyer, Microphase separation 3D printing of binary inorganic polymer precursors to prepare nanostructured carbon-ceramic multimaterials, Adv. Mater. Technol. 9 (13) (2024) 2400337.

[8] J. Wan, B. Fan, S.H. Thang, RAFT-mediated polymerization-induced self-assembly (RAFT-PISA): current status and future directions, Chem. Sci. 13 (15) (2022) 4192–4224.

[9] J. Wang, X. Hu, N. Zhu, K. Guo, Continuous flow photo-RAFT and light-PISA, Chem. Eng. J. 420 (2021) 127663.

[10] N. Li, X.-C. Pan, Controlled radical polymerization: from oxygen inhibition and tolerance to oxygen initiation, Chin. J. Polym. Sci. 39 (9) (2021) 1084–1092.

[11] N. Corrigan, K. Jung, G. Moad, C.J. Hawker, K. Matyjaszewski, C. Boyer, Reversible-deactivation radical polymerization (Controlled/living radical polymerization): From discovery to materials design and applications, Prog. Polym. Sci. 111 (2020) 101311.

[12] F. De Bon, A.B. Barbosa, R.G. Fonseca, M. Fantin, A.C. Serra, J.F.J. Coelho, Large volume and oxygen tolerant photoinduced aqueous atom transfer radical polymerization, Chem. Eng. J. 451 (2023) 138777.

[13] H. Yang, Z. Lu, X. Fu, Q. Li, L. Xiao, Y. Zhao, L. Hou, Visible light-triggered PET-RAFT polymerization by heterogeneous 2D porphyrin-based COF photocatalyst under aqueous condition, Eur. Polym. J. 173 (2022) 111306.

[14] H. Yang, R. Zhao, Z. Lu, L. Xiao, L. Hou, Recognition of "Oxygen-/Water-Fueled" PET-RAFT protocol matched to covalent organic frameworks, ACS Catal. (2023) 2948–2956.

[15] H. Yang, Z. Lu, X. Fu, Q. Li, L. Xiao, R. Zhao, Y. Zhao, L. Hou, Multipath oxygen-mediated PET-RAFT polymerization by a conjugated organic polymer photocatalyst under red LED irradiation, Polym. Chem. 12 (2021) 6998–7004.

[16] G. Ng, S.W. Prescott, A. Postma, G. Moad, C.J. Hawker, C. Boyer, Strategies for achieving oxygen tolerance in reversible addition-fragmentation chain transfer polymerization, Macromol. Chem. Phys. 224 (19) (2023) 2300132.

[17] Q. Ma, X. Zhang, Y. Jiang, J. Lin, B. Graff, S. Hu, J. Lalevee, S. Liao, Organocatalytic PET-RAFT polymerization with a low ppm of organic photocatalyst under visible light, Polym. Chem. (2021) 209–219.

[18] A. Bagheri, C.W.A. Bainbridge, K.E. Engel, G.G. Qiao, J. Xu, C. Boyer, J. Jin, Oxygen tolerant PET-RAFT facilitated 3D printing of polymeric materials under visible LEDs, ACS Appl. Polym. Mater. 2 (2) (2020) 782–790.

[19] Z. Zhang, N. Corrigan, A. Bagheri, J. Jin, C. Boyer, A versatile 3D and 4D printing system through photocontrolled RAFT polymerization, Angew. Chem. Int. Ed. 58 (50) (2019) 17954–17963.

[20] H. Lu, Y. Huang, E. Zhang, Y. Liu, F. Lv, L. Liu, Y. Ma, S. Wang, Photocontrolled RAFT polymerization catalyzed by conjugated polymers under aerobic aqueous conditions, ACS Macro Lett. 10 (8) (2021) 996–1001.

[21] Y. Song, Y. Kim, Y. Noh, V.K. Singh, S. Kumar Behera, A. Abudulimu, K. Chung, R. Wannemacher, J. Gierschner, L. Luer, M.S. Kwon, Organic photocatalyst for ppm-level visible-light-driven reversible addition-fragmentation chain-transfer (RAFT) polymerization with excellent oxygen tolerance, Macromolecules 52 (15) (2019) 5538–5545.

[22] J.H. Lee, H.J. Seo, S.Y. Lee, W.K. Cho, K.-S. Son, On-surface RAFT polymerization using oxygen to form triblock copolymer brushes, Macromol. Chem. Phys. 222 (2022) 2100270.

[23] Z.-H. Zhang, X. Wang, X.-J. Wang, Y. Li, M. Hong, Tris(2,4-difluorophenyl)borane/triisobutylphosphine Lewis pair: A thermostable and air/moisture-tolerant organic catalyst for the living polymerization of acrylates, Macromolecules 54 (18) (2021) 8495–8502.

[24] C.S. Kim, S. Cho, J.H. Lee, W.K. Cho, K.-S. Son, Open-to-air RAFT polymerization on a surface under ambient conditions, Langmuir 36 (39) (2020) 11538–11545.

[25] E.P. Fonseca Parra, B. Chouchene, J.-L. Six, R. Schneider, K. Ferji, Mechanistic insights into oxygen tolerance of graphitic carbon nitride-mediated heterogeneous photoinduced electron transfer-reversible addition fragmentation chain transfer polymerization, ACS Appl. Polym. Mater. 3 (7) (2021) 3649–3658.

[26] Z. Li, S. Ganda, D. Melodia, C. Boyer, R. Chapman, Well-defined polymers for nonchemistry laboratories using oxygen tolerant controlled radical polymerization, J. Chem. Educ. 97 (2) (2020) 549–556.

[27] K. Kaya, B. Kiskan, B. Kumru, B.V.K.J. Schmidt, Y. Yagci, An oxygen-tolerant visible light induced free radical polymerization using mesoporous graphitic carbon nitride, Eur. Polym. J. 122 (2020) 109410.

[28] T. Zhang, J. Yeow, C. Boyer, A cocktail of vitamins for aqueous RAFT polymerization in an open-to-air microtiter plate, Polym. Chem. 10 (34) (2019) 4643–4654.

[29] B. Yuan, T. Huang, X. Wang, Y. Ding, L. Jiang, Y. Zhang, J. Tang, Oxygen-tolerant RAFT polymerization catalyzed by a recyclable biomimetic mineralization enhanceds biological cascade system, Macromol. Rapid Commun. (2021) e2100559.

[30] R.-Y. Li, Z.-S. An, Photoenzymatic RAFT emulsion polymerization with oxygen tolerance, Chin. J. Polym. Sci. 39 (9) (2021) 1138–1145.

[31] W.B. Cai, D.D. Liu, Y. Chen, L. Zhang, J.B. Tan, Enzyme-assisted photoinitiated polymerization-induced self-assembly in continuous flow reactors with oxygen tolerance, Chin. J. Polym. Sci. 39 (9) (2021) 1127–1137.

[32] R. Li, Z. An, Achieving ultrahigh molecular weights with diverse architectures for unconjugated monomers through oxygen-tolerant photoenzymatic RAFT polymerization, Angew. Chem.-Int. Ed. 59 (49) (2020) 22258–22264.

[33] M. Wang, J. Zhang, C. Guerrero-Sanchez, U.S. Schubert, A. Feng, S.H. Thang, Enzyme degassing for oxygen-sensitive reactions in open vessels of an automated parallel synthesizer: RAFT polymerizations, ACS Comb. Sci. 21 (10) (2019) 643–649.

[34] N. Rostami, D. Graf, L. Schranzhofer, S. Hild, T. Hanemann, Overcoming oxygen inhibition effect by TODA in acrylate-based ceramic-filled inks, Prog. Org. Coat. 130 (2019) 221–225.

[35] J. Peng, Q. Xu, Y. Ni, L. Zhang, Z. Cheng, X. Zhu, Visible light controlled aqueous RAFT continuous flow polymerization with oxygen tolerance, Polym. Chem. 10 (16) (2019) 2064–2072.

[36] J. Kirschner, J.M. Becht, J.E. Klee, J. Lalevee, A new phosphine for efficient free radical polymerization under air, Macromol. Rapid Commun. 41 (9) (2020) e2000053.

[37] Y. Shi, H. Gao, L. Lu, Y. Cai, Ultra-fast RAFT polymerisation of poly(ethylene glycol) acrylate in aqueous media under mild visible light radiation at 25 degrees C, Chem. Commun. 11 (2009) 1368–1370.

[38] T.Y. Lee, C.A. Guymon, E.S. Jönsson, C.E. Hoyle, The effect of monomer structure on oxygen inhibition of (meth)acrylates photopolymerization, Polymer 45 (18) (2004) 6155–6162.

[39] T. Haraldsson, M. Johansson, A. Hult, The effects of abstractable hydrogen in radical photopolymerization of maleate/vinyl ether monomers studied with EPR and photo-RTIR, J. Polym. Sci. A Polym. Chem. 48 (13) (2010) 2810–2816.